Andrijana Miković



Full-Stack Software Engineer

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SUMMARY

Full-stack software engineer with 2.5+ years of experience building and deploying an enterprise Web GIS platform used by customers across multiple organizations. Works across the whole stack: Vue.js and TypeScript/JavaScript on the frontend, Node.js and Java REST APIs on the backend, PostgreSQL, SQL Server and Oracle databases, and Docker-based deployments. Initiated and designed the team's in-house UI component library and is contributing to a Java-to-Node.js backend migration. M.Sc. in Software Engineering, University of Belgrade.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, SQL, C, C++
Frontend: Vue.js (Vue 2 & 3), Vuex, Pinia, Vuetify, React, Angular, D3.js, HTML5, CSS3, UI component library design
Backend: Node.js, Express, Java, REST API development, JWT authentication, role-based access control (RBAC)
Databases: PostgreSQL, Microsoft SQL Server, Oracle, MongoDB, database migration
DevOps & Tools: Docker, Apache, IIS (reverse proxy), Windows Server deployment, Git, Jira, Agile
GIS, Data & AI: ArcGIS API for JavaScript, ArcGIS REST services, ArcPy, geoprocessing, LLM integration (Llama), prompt engineering, machine learning, NetworkX, Pandas

PROFESSIONAL EXPERIENCE

Software Engineer I — GDi Solutions Apr 2026 – Present
Enterprise Web GIS platform (GDi Ensemble Smart Portal) · Vue.js, Node.js, PostgreSQL, Docker, Python/ArcPy, Java

- Contributing to the migration of the platform backend from Java to Node.js and consolidating its multi-database layer (SQL Server, Oracle, PostgreSQL) onto PostgreSQL, alongside a full frontend redesign.
- Initiated the team's in-house Vue.js UI component library, a from-scratch replacement for Vuetify: proposed the idea, laid its foundations and core infrastructure, and designed its architecture and components.
- Implemented JWT authentication and role/privilege-based authorization, and built REST API endpoints and business logic used across the platform.
- Containerized platform services with Docker for consistent, repeatable deployments.
- Built an AI-assisted configuration generator integrating a self-hosted Llama LLM with custom prompt engineering tailored to the platform's infrastructure.
- Developed a Python/ArcPy geoprocessing tool that automates spatial data export and analysis.
- Maintain and extend custom features for enterprise customers still on the legacy Java/Vuetify version.

Junior Software Engineer — GDi Solutions Apr 2024 – Apr 2026
Java, Vue.js 2.7, Vuetify, Vuex/Pinia, ArcGIS API, SQL Server, Oracle, PostgreSQL, Apache, IIS

- Developed new platform features and reusable map widgets (file upload, import/export, dynamic forms, tables, filters) with Vue.js, Vuetify and the ArcGIS API for JavaScript.
- Implemented Java backend functionality — REST endpoints, database access and proxy logic — and added Oracle and PostgreSQL support alongside Microsoft SQL Server.
- Deployed the platform to Windows client servers for multiple enterprise customers: configured Apache, set up IIS as a reverse proxy, and migrated customer databases between environments.
- Handled production debugging and customer support; helped define the architecture of new modules and managed state with Vuex and Pinia.

Undergraduate Teaching Assistant — School of Electrical Engineering, University of Belgrade Oct 2023 – Feb 2024

- Assisted in machine learning labs (Python), guiding students through practical assignments and grading submissions.

Software Engineer Intern — Microsoft Development Center Serbia Mar 2022 – May 2022

- Developed frontend features with React.js and D3.js on the Maps team's web application for viewing and editing map data (C# backend), integrating external point-of-interest data.

EDUCATION

M.Sc. Software Engineering 2023 – 2025

University of Belgrade, School of Electrical Engineering

Focus: big data processing, secure software development and cyber security, UI programming, social network analysis.

Thesis: Analysis of trends in world cinematography in the early 21st century (Aug 2025) — network and statistical analysis of ~1.8M titles (see Projects).

B.Sc. Computer Engineering and Information Technology 2019 – 2023

University of Belgrade, School of Electrical Engineering

Core coursework: algorithms and data structures, object-oriented programming (C++, Java), operating systems, system software, compiler construction, computer architecture, web programming, intelligent systems, machine learning, VLSI design.

Thesis: Machine learning for diabetes detection (Sep 2023) — compared KNN, logistic regression and decision trees on 100,000 samples, handling class imbalance with under-sampling and ADASYN.

SELECTED PROJECTS

Master's Thesis: Collaboration Networks in World Cinema — *Python*, *NetworkX*, *igraph*, *Gephi* 2025

Collected film data via the TMDB API and Rotten Tomatoes scraping, then modeled ~1.8M IMDb titles (2001–2024) as collaboration networks of actors, directors and production companies; analyzed centrality, community structure, clustering and assortativity, combined with statistical trend analysis.

Cake Shop Web Application — *Vue 3*, *Vuetify 3*, *Vuex*, *Vue Router*, *Firebase* 2024

Single-page app with separate customer and worker roles: customers browse the catalog, manage a cart and place orders; workers add products and staff and accept or reject orders with a preparation time or reason. Route guards protect authenticated and worker-only pages; data stored in Firebase Realtime Database.

Web Application Security Hardening — *Java*, *Spring Boot*, *Spring Security* 2024

Audited a Spring Boot web shop and fixed its vulnerabilities: SQL injection, XSS and CSRF (including a working CSRF exploit to demonstrate the issue), and added application logging and an audit trail of user actions.

Library Web Application — *Angular*, *TypeScript*, *Node.js*, *Express*, *MongoDB* 2022

Full-stack MEAN application with reader and admin roles: registration with admin approval, user blocking, basic and advanced book search, borrowing with history, ratings and comments, and book requests.

Systems Programming: Compiler, Toolchain & Kernel — *Java*, *C++* 2021 – 2023

Micro-Java compiler with JFlex/CUP lexical and syntax analysis; assembler, linker and emulator for an abstract CPU; multi-threaded kernel with time-sharing, context switching and semaphores for x86 and RISC-V.

More projects (machine learning, NLP, AI search, hardware) on GitHub and at andrijanamikovic.github.io.

ADDITIONAL

Languages: Serbian (native), English (professional working proficiency)

Achievements: Petnica Science Center — applied physics & electronics programs (2017–2019); led a school applied physics & electronics club; exhibited at the Belgrade Science Fair (3 years) and Vienna Maker Faire.